

## Section 5.2

24. Let  $h(x,y)$  denote the number of individuals between A and B. Then the table for  $h(x,y)$

| $y \backslash x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|------------------|---|---|---|---|---|---|
| 1                | X | 2 | 3 | 4 | 3 | 2 |
| 2                | 2 | X | 2 | 3 | 4 | 3 |
| 3                | 3 | 2 | X | 2 | 3 | 4 |
| 4                | 4 | 3 | 2 | X | 2 | 3 |
| 5                | 3 | 4 | 3 | 2 | X | 2 |
| 6                | 2 | 3 | 4 | 3 | 2 | X |

For each situation,  $p(x,y) = \frac{1}{30}$ .

$$\begin{aligned}
 \text{Thus } E(h(x,y)) &= \sum_x \sum_y h(x,y) \cdot p(x,y) \\
 &= \frac{1}{30} (2+3+4+3+\dots+2) \\
 &= 2.8
 \end{aligned}$$

26. Let  $h(x,y)$  denote the revenue.  $h(x,y) = 3x + 10y$ .

$$\begin{aligned}
 E(h(x,y)) &= \sum_x \sum_y h(x,y) \cdot p(x,y) \\
 &= (0 \times 3 + 0 \times 10) \times 0.025 + (0 \times 3 + 1 \times 10) \times 0.015 + \dots + \\
 &\quad (5 \times 3 + 2 \times 10) \times 0.020 \\
 &= 15.4
 \end{aligned}$$

33. When  $X$  and  $Y$  are independent,  $E(XY) = E(X)E(Y)$

$$\text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

$$= E(X) \cdot E(Y) - E(X) \cdot E(Y)$$

$$= 0$$

$$\text{and } \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y} = 0$$

35. a.  $\text{Cov}(aX+b, cY+d)$

$$= E[(aX+b)(cY+d)] - E(aX+b) \cdot E(cY+d)$$

$$= E(acXY + adX + bcY + bd) - (aE(X) + b)(cE(Y) + d)$$

$$= acE(XY) + adE(X) + bcE(Y) + bd - (acE(X)E(Y) + adE(X) + bcE(Y) + bd)$$

$$= acE(XY) - acE(X)E(Y)$$

$$= ac \text{Cov}(X, Y)$$

b.  $\text{Corr}(aX+b, cY+d) = \frac{\text{Cov}(aX+b, cY+d)}{\sigma_{aX+b} \cdot \sigma_{cY+d}}$

$$= \frac{a \cdot c \text{Cov}(X, Y)}{\sqrt{a^2 \sigma_X^2} \cdot \sqrt{c^2 \sigma_Y^2}}$$

$$= \frac{ac \text{Cov}(X, Y)}{|ac| \cdot \sigma_X \cdot \sigma_Y}$$

when  $a$  and  $c$  have the same sign,

$|ac| = ac$ . Thus

$$\begin{aligned}\text{Corr}(aX+b, cY+d) &= \frac{ac \text{Cov}(X, Y)}{ac \cdot \sigma_X \sigma_Y} \\ &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \\ &= \text{Corr}(X, Y)\end{aligned}$$

c. If  $a$  and  $c$  have opposite sign,  $|ac| = -ac$

$$\begin{aligned}\text{Thus } \text{Corr}(aX+b, cY+d) &= \frac{ac \text{Cov}(X, Y)}{-ac \sigma_X \sigma_Y} \\ &= -\text{Corr}(X, Y)\end{aligned}$$

## Section 5.3

38. a. For each  $T_0$ ,  $p(t) = p(x_1) \cdot p(x_2)$ .

and all possible  $t = 0, 1, 2, 3, 4$

when  $t=0$ ,  $p(t=0) = 0.2 \times 0.2 = 0.04$

when  $t=1$ ,  $p(t=1) = 0.2 \times 0.5 + 0.5 \times 0.2 = 0.2$

when  $t=2$ ,  $p(t=2) = 0.2 \times 0.3 \times 2 + 0.5 \times 0.5 = 0.37$

when  $t=3$ ,  $p(t=3) = 0.5 \times 0.3 \times 2 = 0.3$

when  $t=4$ ,  $p(t=4) = 0.3 \times 0.3 = 0.09$

The pmf:

| $t$    | 0    | 1   | 2    | 3   | 4    |
|--------|------|-----|------|-----|------|
| $p(t)$ | 0.04 | 0.2 | 0.37 | 0.3 | 0.09 |

$$\begin{aligned} \text{b. } \mu_{T_0} &= \sum t \cdot p(t) = 0.04 \times 0 + 0.2 \times 1 + \dots + 0.09 \times 4 \\ &= 2.2 \end{aligned}$$

And  $\mu_{T_0} = 2\mu$ . That is, the expected value of  $T_0$  is twice of  $\mu$ , the population mean.

$$\text{c. } \sigma_{T_0}^2 = E(t^2) - [E(t)]^2$$

$$\begin{aligned} E(t^2) &= 0 \times 0.04 + 1 \times 0.2 + 4 \times 0.37 + 9 \times 0.3 + 16 \times 0.09 \\ &= 5.82 \end{aligned}$$

$$[E(t)]^2 = 4.84$$

$$\text{Thus } \sigma_{T_0}^2 = 0.98$$

And  $\sigma_{T_0}^2 = 2\sigma^2$ . That is, the variance of  $T_0$  is twice of  $\sigma^2$ , the population variance.



d. Since the pmf of  $X_1, X_2, X_3$  and  $X_4$  are the same,  $T = X_1 + X_2 + X_3 + X_4$

$$\begin{aligned} E(T_0) &= E(X_1) + E(X_2) + E(X_3) + E(X_4) \\ &= 4\mu = 4.4 \end{aligned}$$

$$\begin{aligned} V(T_0) &= V(X_1) + V(X_2) + V(X_3) + V(X_4) \\ &= 4\sigma^2 = 1.96 \end{aligned}$$

e. when  $T_0 = 8$ , all  $X$  takes 2.

$$\text{Thus } P(T_0 = 8) = (0.3)^4 = 0.0081$$

when  $T_0 = 7$ , there exists one  $X$  takes 1 and the rest takes 2.

$$\text{Thus } P(T_0 = 7) = \binom{4}{1} \cdot 0.5 \times 0.3^3 = 0.054$$

$$\text{Thus } P(T_0 \geq 7) = P(T_0 = 8) + P(T_0 = 7) = 0.0621$$

41. a.  $\bar{X} = \frac{X_1 + X_2}{2}$ . The possible values are 1, 1.5, 2, 2.5, 3, 3.5, 4

when  $\bar{X} = 1$ ,  $P = 0.4 \times 0.4 = 0.16$

when  $\bar{X} = 1.5$ ,  $P = \binom{2}{1} \times 0.4 \times 0.3 = 0.24$

when  $\bar{X} = 2$ ,  $P = 0.3 \times 0.3 + \binom{2}{1} \times 0.4 \times 0.2 = 0.25$

when  $\bar{X} = 2.5$ ,  $P = \binom{2}{1} \times 0.3 \times 0.2 + \binom{2}{1} \times 0.4 \times 0.1 = 0.20$

when  $\bar{X} = 3$ ,  $P = 0.2 \times 0.2 + \binom{2}{1} \times 0.3 \times 0.1 = 0.10$

when  $\bar{X} = 3.5$ ,  $P = \binom{2}{1} \times 0.2 \times 0.1 = 0.04$

when  $\bar{X} = 4$ ,  $P = 0.1 \times 0.1 = 0.01$

thus the probability distribution of  $\bar{X}$ :

| $\bar{X}$    | 1    | 1.5  | 2    | 2.5  | 3    | 3.5  | 4    |
|--------------|------|------|------|------|------|------|------|
| $P(\bar{X})$ | 0.16 | 0.24 | 0.25 | 0.20 | 0.10 | 0.04 | 0.01 |

b.  $P(\bar{X} \leq 2.5) = 0.16 + 0.24 + 0.25 + 0.20 = 0.85$

c. All possible outcomes of  $R$  are 0, 1, 2, 3

when  $R = 0$ ,  $P = 0.4 \times 0.4 + 0.3 \times 0.3 + 0.2 \times 0.2 + 0.1 \times 0.1 = 0.30$

when  $R = 1$ ,  $P = \binom{2}{1} \times 0.4 \times 0.3 + \binom{2}{1} \times 0.3 \times 0.2 + \binom{2}{1} \times 0.2 \times 0.1 = 0.40$

when  $R = 2$ ,  $P = \binom{2}{1} \times 0.4 \times 0.2 + \binom{2}{1} \times 0.3 \times 0.1 = 0.22$

when  $R = 3$ ,  $P = \binom{2}{1} \times 0.4 \times 0.1 = 0.08$

Thus the distribution of  $R$  is:

| $r$    | 0    | 1    | 2    | 3    |
|--------|------|------|------|------|
| $p(r)$ | 0.30 | 0.40 | 0.22 | 0.08 |

d.  $\bar{X} = \frac{x_1 + x_2 + x_3 + x_4}{4}$ . For  $\bar{x} \leq 1.5$ , possible  $\bar{x} = 1, 1.25, 1.5$

when  $\bar{x} = 1$ .  $p = (0.4)^4 = 0.0256$

when  $\bar{x} = 1.25$ , there exists one  $x_i = 2$ .

$$p = \binom{4}{1} \cdot (0.4)^3 \cdot 0.3 = 0.0768$$

when  $\bar{x} = 1.5$ , there exist two  $x_i = 2$  or one  $x_i = 3$ .

$$p = \binom{4}{2} \cdot (0.4)^2 \cdot (0.3)^2 + \binom{4}{1} \cdot (0.4)^3 \cdot 0.2 = 0.1376.$$

Thus  $p(\bar{x} \leq 1.5) = 0.0256 + 0.0768 + 0.1376 = 0.24$

